

Department of Computer Science & Engineering (Data Science)

Synopsis

On

SMART IMAGE REALITY CHECKER

1. INTRODUCTION

1.1 Overview

The Smart image reality checker System is a deep-learning-driven software solution designed to analyze digital media, extract visual features, and build classification models to predict whether an image is a real photograph or generated by Artificial Intelligence. This project helps in understanding visual artifacts, pixel-level discrepancies, and synthetic textures using convolutional neural networks (CNNs).

1.2 Purpose

The purpose of this project is to analyze digital images to uncover hidden AI artifacts and use deep learning techniques to classify image authenticity with high confidence. In today's rapidly evolving digital landscape, generative AI models (like Midjourney and DALL-E) create vast amounts of hyper-realistic synthetic images. However, this technology is increasingly used to spread misinformation and deepfakes. Through this project, the aim is to transform complex image data into clear authenticity predictions by applying computer vision preprocessing, feature extraction, and transfer learning. This system will help users and organizations make informed decisions about media credibility. Ultimately, the project intends to build a smart, lightweight solution that not only detects AI generation but also enhances trust in digital platforms.

2. LITERATURE SURVEY

2.1 Existing Problem

Although generative AI platforms produce millions of images, most standard verification tools do not fully utilize deep learning for detection purposes. Traditional image analysis systems only focus on metadata extraction or reverse image searches, without offering deeper insights into pixel-level inconsistencies or AI-generated noise patterns. Many fact-checkers and average users still rely on visual inspection, which is time-consuming, error-prone, and provides no mathematical confidence. There is no lightweight automated system readily available to analyze image textures, predict generation sources, or flag deepfakes efficiently on standard hardware. Additionally, without proper UI tools, verifying images across social media becomes difficult. As a result, people often consume media based on assumptions rather than verified authenticity, leading to the spread of misinformation, compromised digital trust, and potential reputational damage.

2.2 Proposed Solution

To overcome these challenges, this project proposes a complete deep-learning-driven Smart image reality checker System using Python, computer vision, and transfer learning. The system will collect historical and synthetic image data, either from real-world sources like the Real vs AI Generated images dataset (Kaggle) or self-generated datasets. The data will go through preprocessing steps such as resizing, normalizing pixel values, and color space conversion using OpenCV. Exploratory Data Analysis (EDA) will be performed to identify visual patterns in real vs. fake images. For the predictive part, pre-trained Convolutional Neural Networks (CNNs) like MobileNetV2 will be fine-tuned to predict image authenticity based on learned spatial features. The system will feature an interactive Streamlit UI where users can upload images and instantly receive a confidence score. This solution will help individuals and businesses make smarter, verified decisions and gain a reliable defense against digital forgery.

3. THEORETICAL ANALYSIS

3.1 Block Diagram

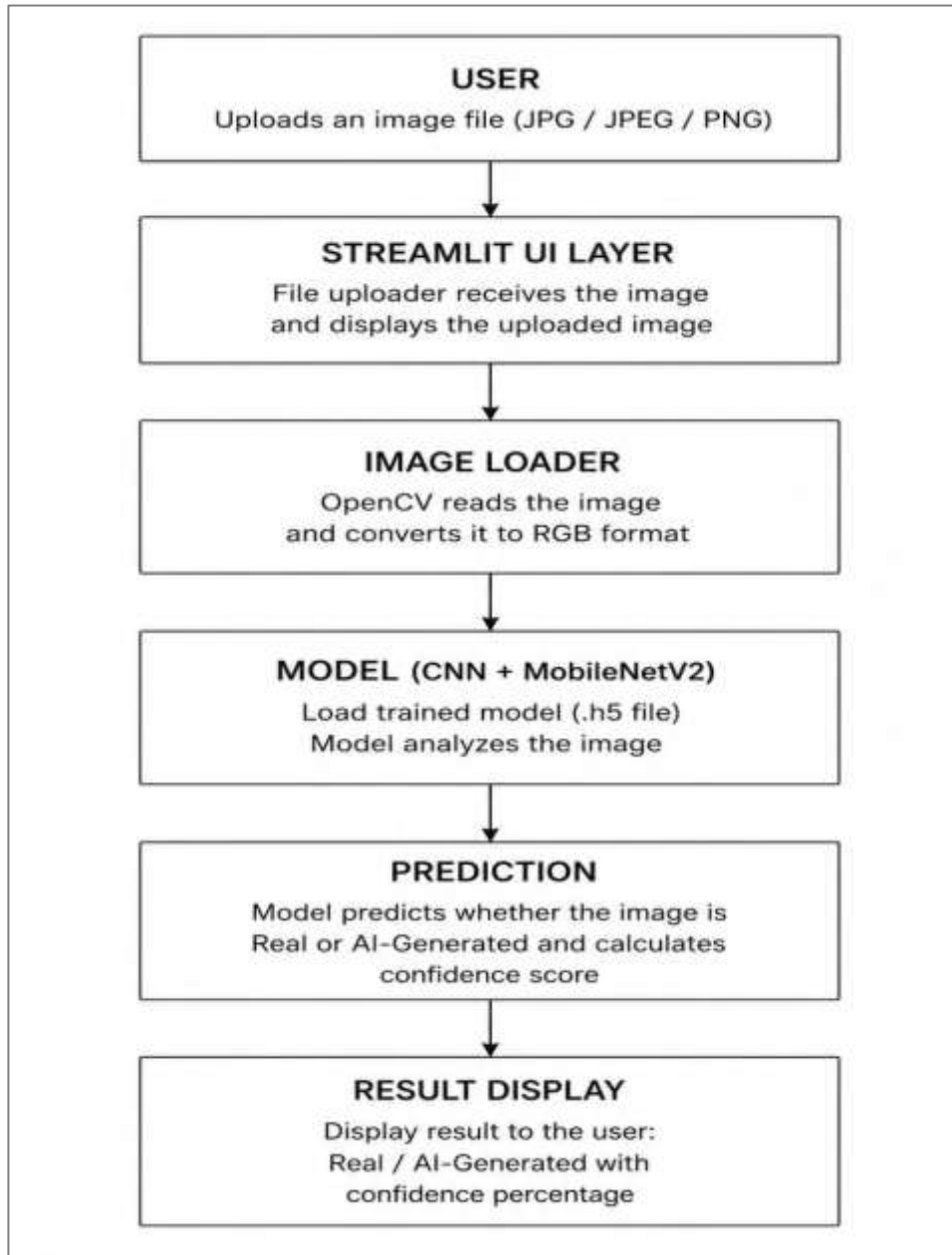


Fig.1. Workflow of SMART IMAGE REALITY CHECKER

3.2 Hardware/Software Designing

Hardware Requirements:

- Processor: Intel i3/i5/i7 or Ryzen 3/5
- RAM: 8GB
- Storage: 128 GB HDD or more
- Network: 10/100 Mbps Internet Connection
- OS: Linux-Based or Windows Server

Software Requirements:

- Programming Language: Python
- Libraries: NumPy, Pandas, Pillow
- ML frame works: TensorFlow / Keras (MobileNetV2)
- Visualization: Matplotlib / Streamlit
- IDE / Tools: Jupyter Notebook / VS Code / Google Colab
- Dataset: Real vs AI Generated Images Dataset
- Version Control: Git with GitHub

4. APPLICATIONS

- **Social Media Verification and Misinformation Prevention**

Platforms can predict and flag AI-generated images using structural data trends. This helps in preventing the spread of fake news, deepfakes, and manipulated media campaigns.

- **Journalism and Fact-Checking**

By predicting image authenticity, news agencies can maintain credibility, preventing the publication of synthetic media and ensuring proper source validation.

- **Copyright and Intellectual Property Protection**

The system helps in analyzing digital artwork to detect if AI was used in its creation. This allows platforms to enforce policies regarding AI-generated content and protect human artists.

- **Academic and Professional Integrity**

Image uploads in academic research, digital portfolios, or identity verification portals can be tracked and validated. This helps identify fabricated data or stolen synthetic assets.

- **Content Moderation Optimization**

By studying the influx of fake images, automated moderation systems can determine the best filtering thresholds that increase platform safety without slowing down user uploads.

REFERENCES

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- [3] Wang, S., et al. (2020). “CNN-generated images are surprisingly easy to spot... for now” — IEEE Conference on Computer Vision and Pattern Recognition (CVPR).
- [4] Rossler, A., et al. (2019). “FaceForensics++: Learning to Detect Manipulated Facial Images” — International Conference on Computer Vision (ICCV).
- [5] Li, Y., et al. (2025). "Bridging the Gap Between Ideal and Real-world Evaluation: Benchmarking AI-Generated Image Detection." — *International Conference on Computer Vision (ICCV)*

GitHub ID:

<https://github.com/hariiom08/Smart-Image-Reality-Checker.git>

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